$|$

$\frac{e^{-E_\theta(\mathbf{o},\mathbf{a})}}{Z(\mathbf{o},\theta)},$

$Z(\mathbf{o},\theta)$

$\mathbf{a}$

$\mathbf{x}_K \sim \mathcal{N}(0, I)$

$_1$, $\mathbf{x}_0$,

$\mathbf{x}_0$

$^{k-1}$

$\alpha_k\big(\mathbf{x}k - \gamma_k\,\varepsilon\theta(\mathbf{x}_k, k)\big)$

$\sigma_k^2 I),$

$\varepsilon_\theta k$

$\alpha_k, \gamma_k, \sigma_k$

$\eta\nabla_\theta(\mathbf{x}),$

$-\nabla_{\mathbf{a}} E_\theta$

ε_θ

tT_o

$\mathbf{O}_t T_a$

$\mathbf{A}_t$

$\mathbf{O}_t$

$\mathbf{A}_t^K \sim \mathcal{N}(0, I)$

$\varepsilon_\theta K$

$\mathbf{A}_t^0$

$\mathbf{O}_t$